Primary Results of a Sensitivity Analysis of Forest Structural Characteristics Using ESA BIOMASS P-Band Data: A Case Study of Tropical Montane and Savanna Ecosystems in Kenya

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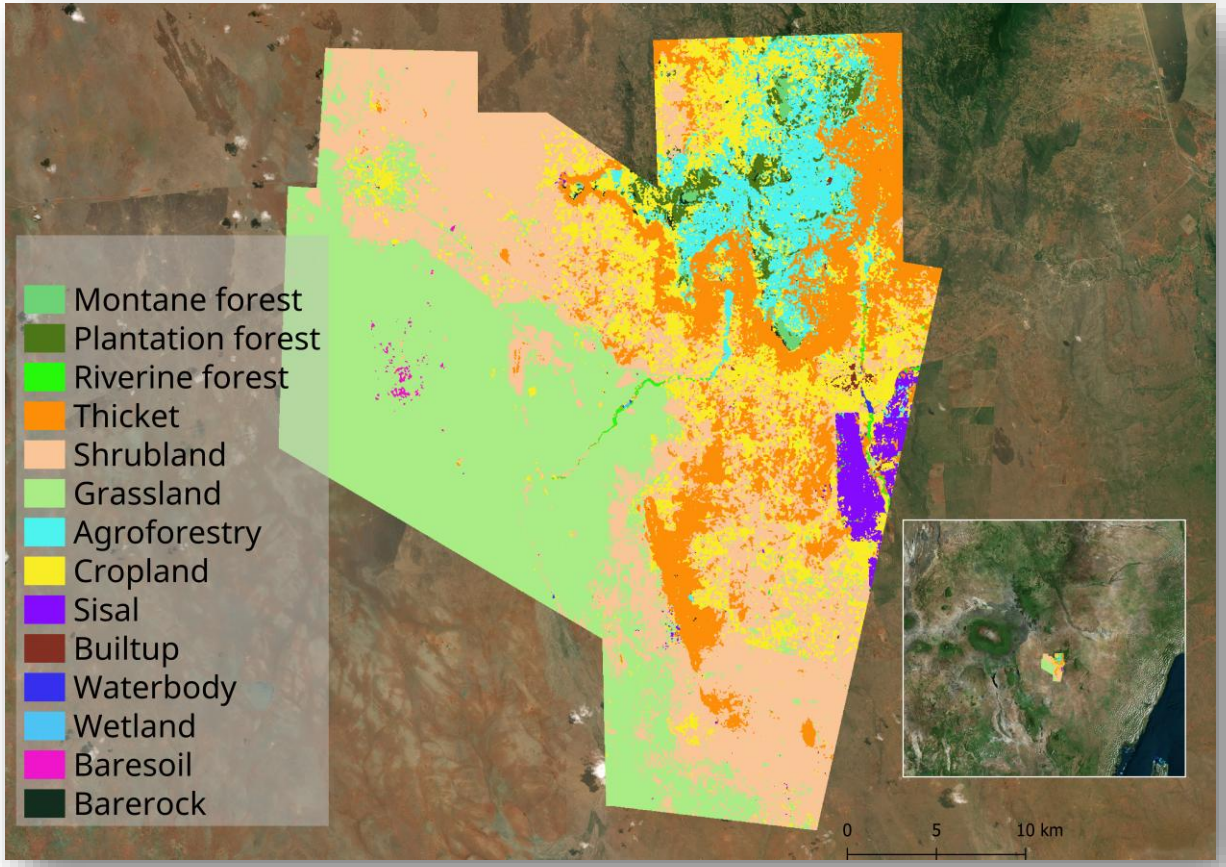
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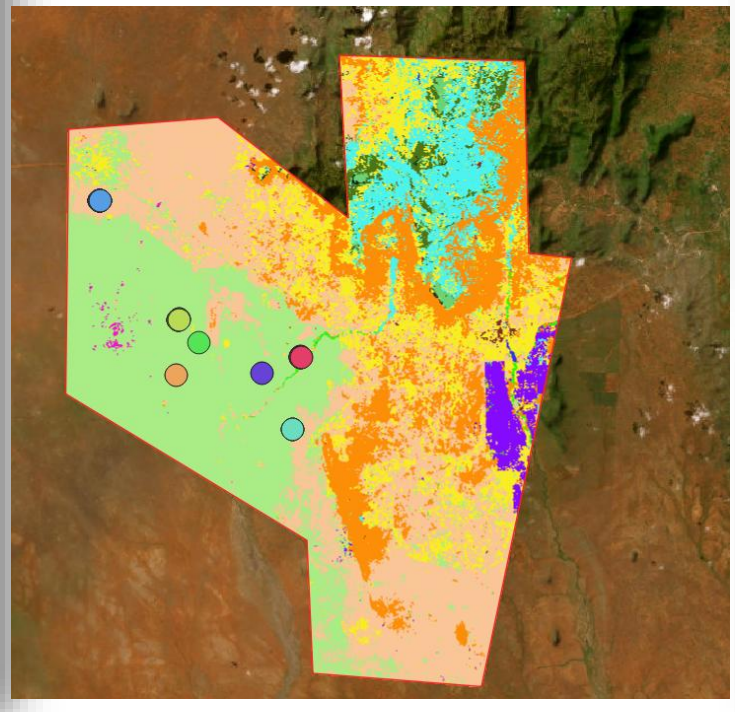
[5]Finnish Meteorological Institute, Finland

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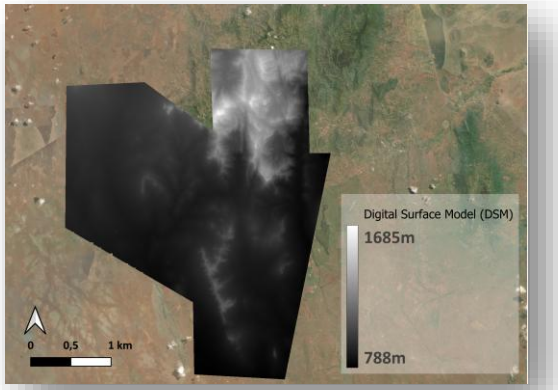
Case study and Data

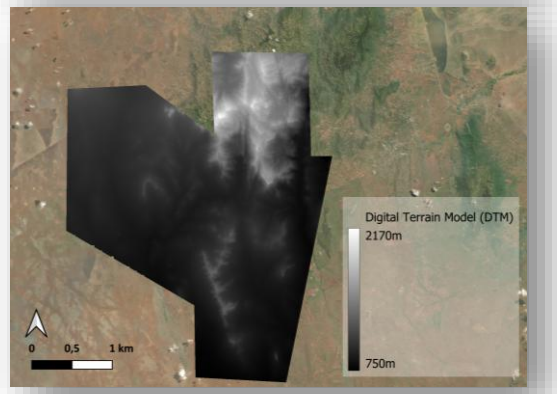


Field Measurements



Airborne LiDAR



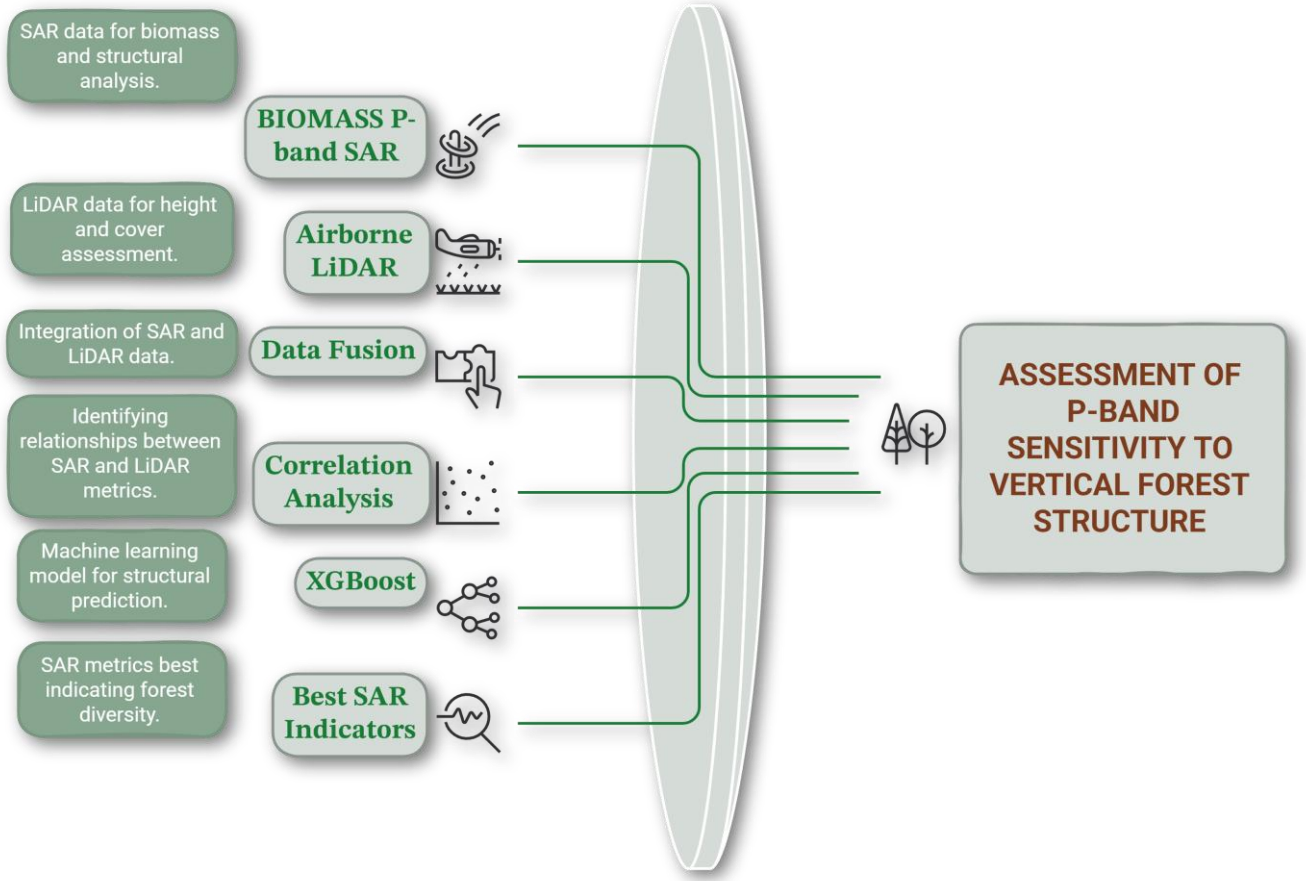


BIOMASS SAR data		
L1a	SCS	11.01.2026

P. K. E. Pellikka *et al.*, "Agricultural Expansion and Its Consequences in the Taita Hills, Kenya," in *Developments in Earth Surface Processes*, vol. 16, Elsevier, 2013, pp. 165–179. doi: 10.1016/B978-0-444-59559-1.00013-X.

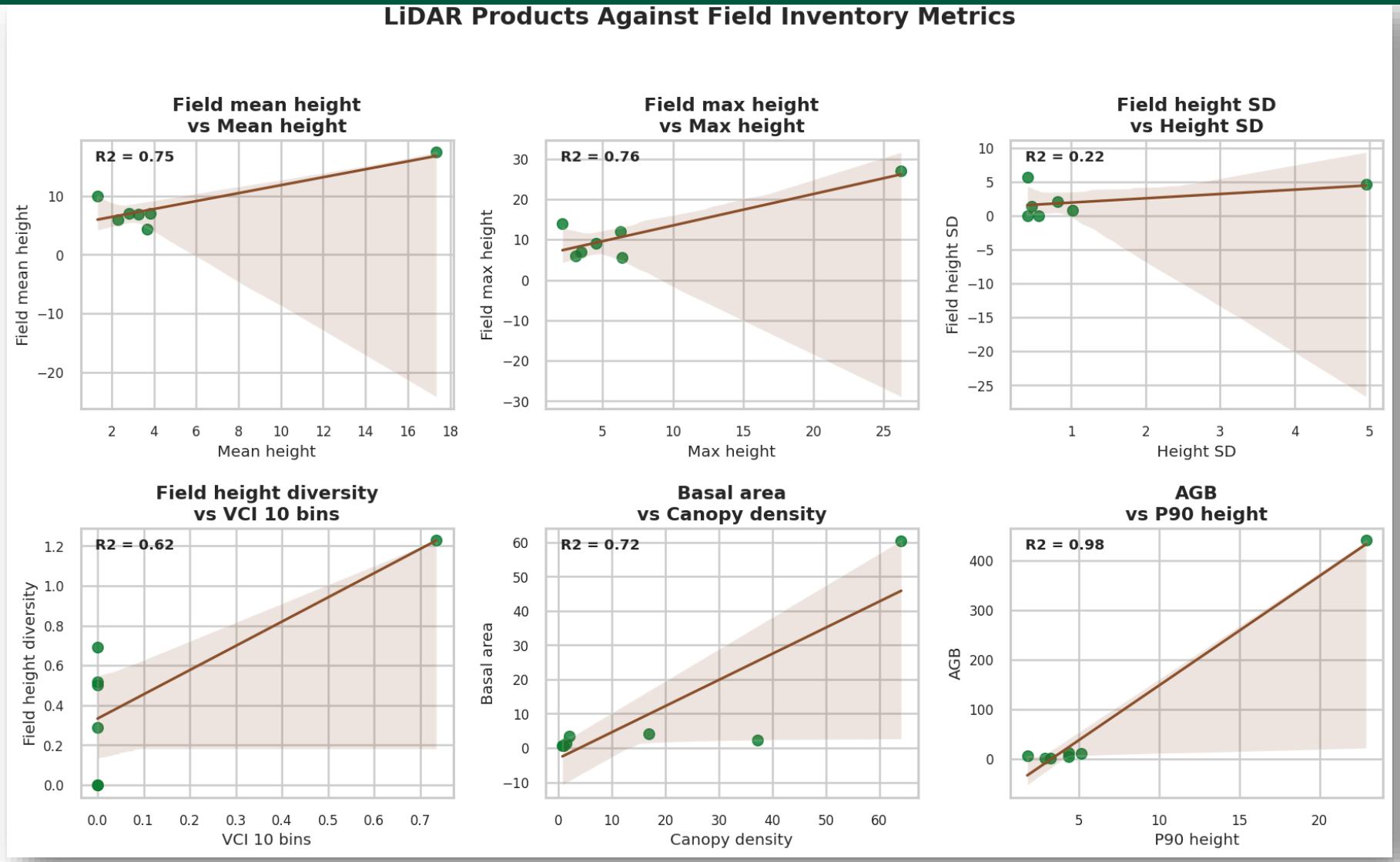
Objectives & Methodology

- Assessing the potential of P-band BIOMASS SAR data for predicting LiDAR-derived forest structural metrics.
- Evaluating the predictability of canopy height, vertical complexity, canopy density and height variability, etc. using Machine Learning (XGBoost regression).
- Identifying the most informative SAR polarimetric and texture features for forest structure estimation, within the study area.

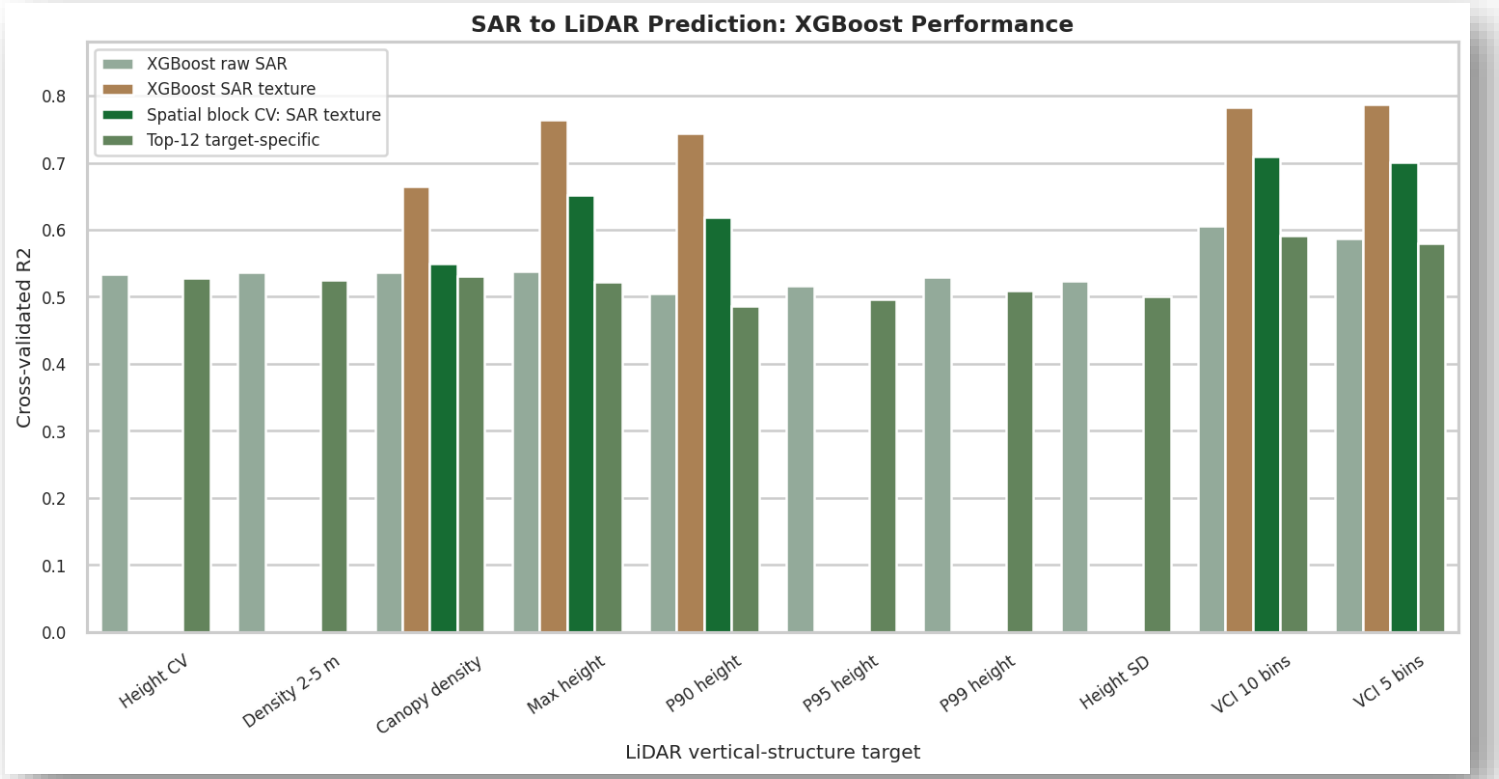


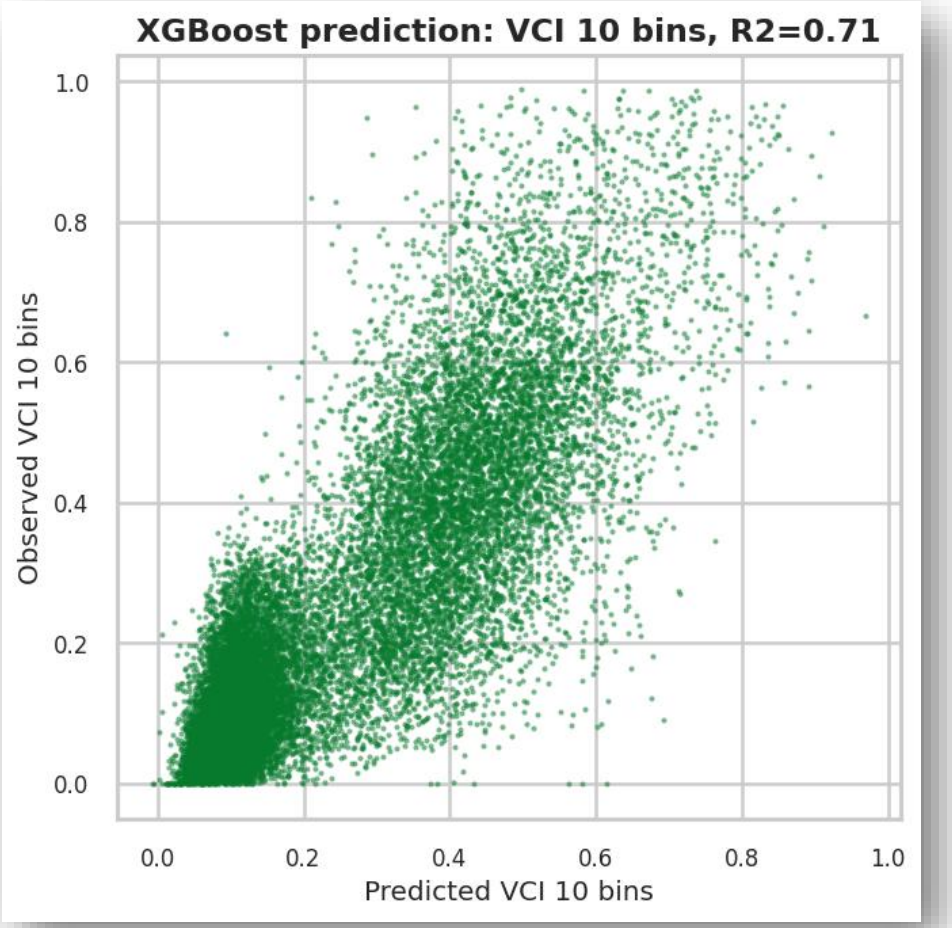
Avoiani, Elizaveta, et al. "Forest Diversity Characterization Across Tropical Montane and Savanna Ecosystems Using Spectral and ESA BIOMASS P-Band Observations." *Fringe 2026*. 2026.

LiDAR vs. Field Observation

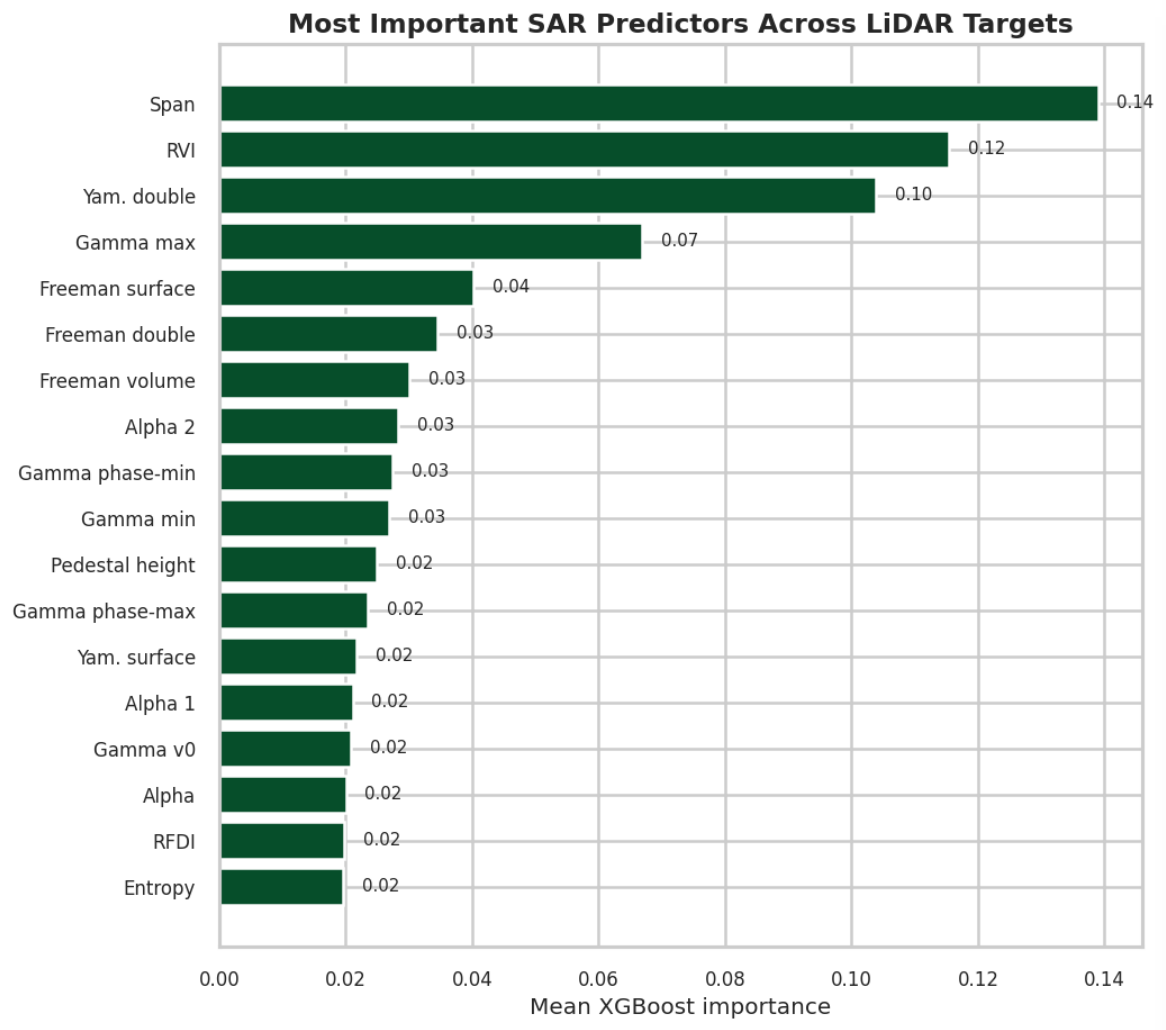


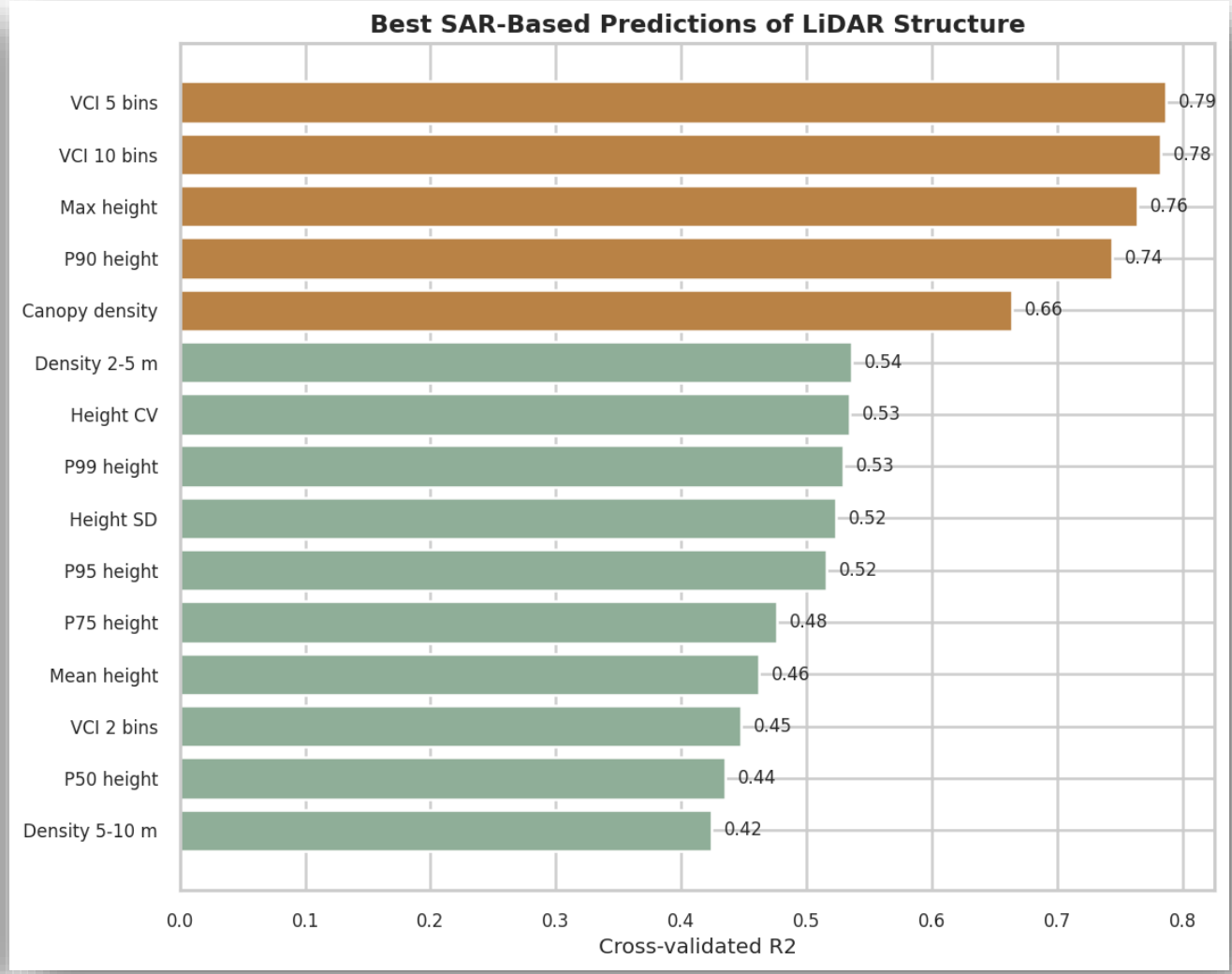
Spatial Block Cross Validation



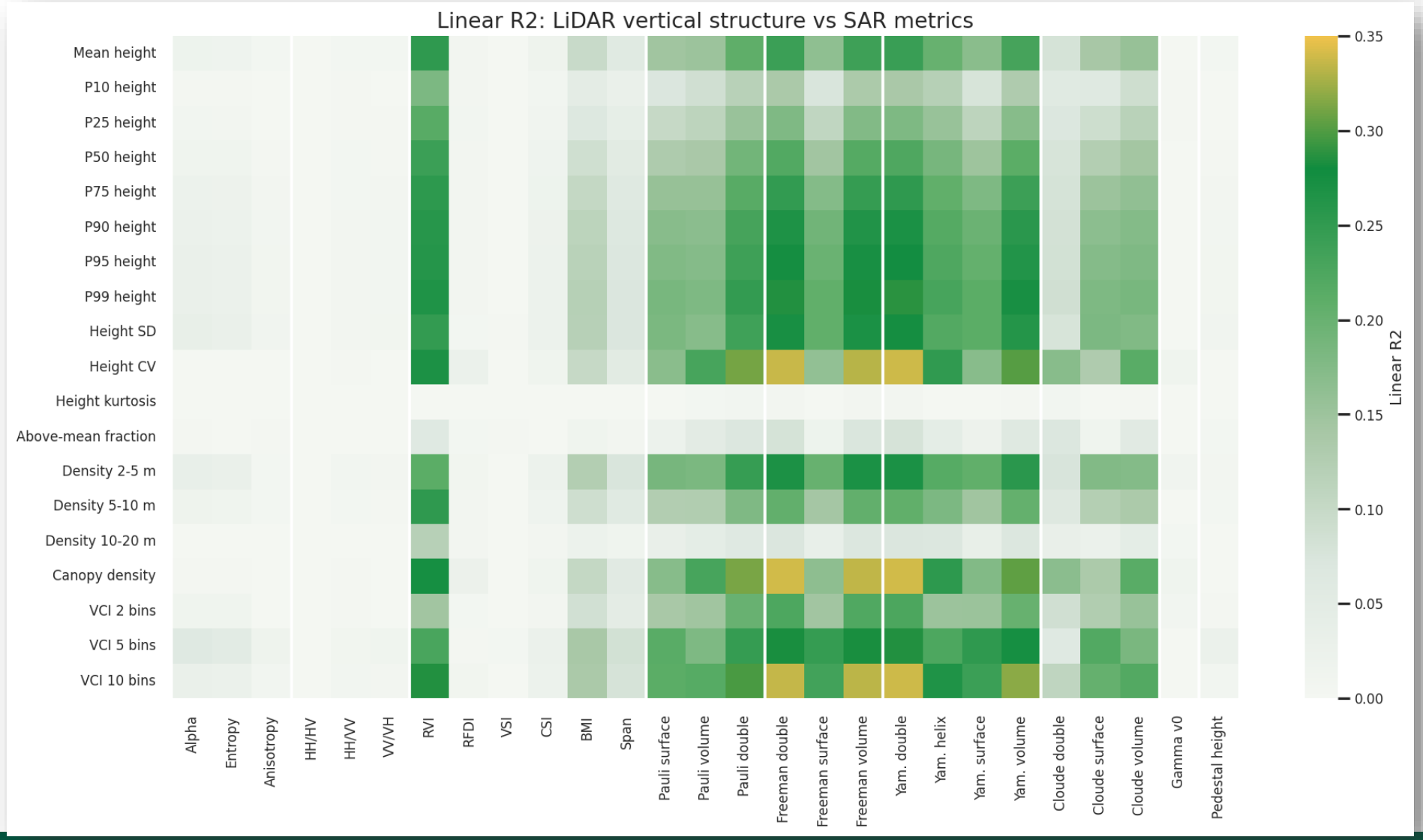


Important SAR Predictors





LiDAR derived Vertical Structure Compared with P-band SAR metrics



Conclusion

- The study demonstrates the potential of P-band SAR polarimetric and texture features, combined with XGBoost, to predict LiDAR-derived forest structural metrics.

- However, the predictive performance varies across metrics, with the strongest results obtained for vertical complexity indices (VCI; (R^2 = 0.78–0.79) and upper-canopy height metrics, including maximum height (R^2 = 0.76) and P90 height (R^2 = 0.74).

- Polarimetric features provide valuable information, with Span, RVI and Yamaguchi double-bounce scattering showing the highest mean feature importance.

- Spatial block cross-validation yields lower performance for several targets, highlighting the need for independent validation.

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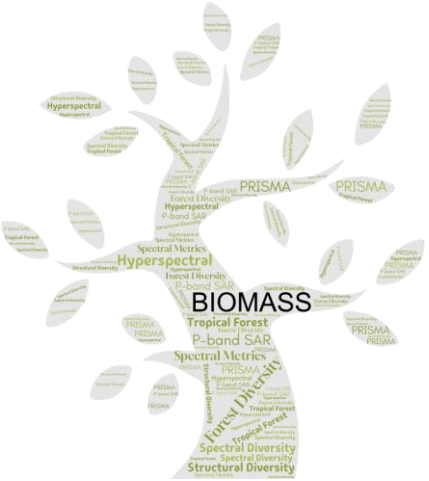
EXDIGIT

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Thank you for your attention!

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